

Problem Statement – AI-Powered Nutrition Agent

Why Personalized Nutrition Guidance is a Critical Need in 2025

The Challenge: A Global Nutrition Crisis

Good nutrition is the foundation of health, but individuals often struggle with **diet planning, portion control, and tracking nutritional intake**. Information about healthy diets, disease-specific nutrition, and food composition is scattered across unreliable sources. Without **personalized, real-time guidance**, many fail to maintain balanced diets or make informed food choices.

The Objective: Intelligent, Personalized Nutrition at Scale

Build an **AI-powered Nutrition Agent** that delivers **personalized dietary insights and guidance** for individuals, families, and health professionals. The solution should:

Personalized Diet Plans – Recommend meals based on age, health conditions, allergies, cultural preferences, and fitness goals.

Food Data RAG Retrieval – Fetch and summarize nutritional values (calories, macros, vitamins) from reliable sources.

Diet Tracking & Feedback – Allow users to log meals via text, images, or voice, and provide instant nutritional analysis.

Preventive Health Focus – Suggest diets for chronic disease management (e.g., diabetes-friendly or heart-healthy plans).